DATA VISUALIZATION AND ANALYTICS

DEFINITION

1. Data

- **Data** refers to raw facts, figures, symbols, or observations collected from various sources.
- **Explanation**
- Data by itself has no meaning until it is processed, analyzed, and interpreted.
- **Examples**
- Numbers: 85, 1200, 45.6
- Text: "Student", "Sales Report"
- Images, videos, sensor readings

QUESTIONS

- **1.What is data and why is it called "raw facts"?**
- Data is called raw facts because it is unprocessed and has no meaning by itself until it is analyzed or interpreted.
- **2. Can data exist without information? Explain.**
- Yes, data can exist without information, but information cannot exist without data. Data becomes information only after processing.
- **3. Give examples of data from daily life.**
- Exam marks, temperature readings, mobile call logs, and bank transactions.
- **4.Why is data considered an important asset today?**
- Because data helps organizations make decisions, predict trends, and improve services.

- **5. How does data help in decision-making?**
- Data provides evidence and insights that reduce guesswork and improve accuracy in decisions.
- **6. What problems can occur if metadata is missing?**
- **Answer:**
  Data may be misunderstood, misused, or become difficult to analyze.

METADATA

Metadata is **data about data**. It provides information that describes, explains, or gives context to data.

Explanation

- Metadata helps users understand:
- What the data represents
- How and when it was created
- How it should be used

Examples

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- Column name: *Age*
- Data type: Integer
- Date created: 10-Jan-2026
- File size and format

5 V'S OF DATA

1. Volume

- **Meaning:**

  The **amount of data** generated and stored.

- **Explanation:**

  Data today is generated in huge quantities from social media, sensors, transactions, and devices.

- **Example:**

Facebook posts, YouTube videos, transaction records

- Data measured in **terabytes, petabytes, or exabytes**

2. VALUE

- **Meaning:**

The **usefulness of data in generating insights or business benefits**.

Not all data is valuable; value comes from meaningful analysis.

- **Example:**
- Customer behavior analysis for marketing
- Fraud detection in banking
- Predictive maintenance in industry

3. VELOCITY

- **Meaning:**
  The **speed at which data is generated, collected, and processed**.
- **Explanation:**
  Data often arrives continuously and needs real-time or near real-time processing.
- **Example:**
- Live stock market data
- Online payment transactions
- IoT sensor data

4. VARIETY

- **Meaning:**
  The different types and formats of data.
- **Explanation:**
  Data is no longer only structured; it also includes semi-structured and unstructured data.
- **Types:**
- Structured: Tables, databases
- Semi-structured: XML, JSON
- Unstructured: Text, images, videos
- **Example:**
- Emails, images, audio files, social media posts

5. VERACITY

- **Meaning**
  The **quality, accuracy, and reliability of data**.
- **Explanation:**
  Data may be incomplete, inconsistent, or noisy, affecting analysis results.
- **Example:**
- Duplicate customer records
- Missing sensor readings
- False information on social media

TYPES OF DATA BASED ON STRUCTURE

1. Structured Data

- **Definition:**
  Data that is organized in a fixed format, usually in rows and columns.
- **Characteristics:**
- Easily stored in databases
- Easy to search and analyze
- **Examples:**
- Student records
- Sales tables
- Employee databases

2. SEMI-STRUCTURED DATA

- **Definition:**
  Data that does not follow a rigid structure but has some organizational markers.
- **Characteristics:**
- Uses tags or labels
- Flexible structure
- **Examples:**
- XML files
- JSON files
- Emails

3. UNSTRUCTURED DATA

Data with no predefined format.

- **Characteristics:**
- Difficult to analyze directly
- Large volume
- **Examples:**
- Images
- Videos
- Audio files
- Social media posts

WHAT INFORMATION ABOUT YOU CAN BE COLLECTED?

- Age, Gender, Marks, Height, Favorite subject, Phone brand, Monthly expenses etc.
- “All this information is **data**, and today we will learn how data is classified.”
- **Data is mainly of two types:**
- **Qualitative (Categorical) Data**
- **Quantitative (Numerical) Data**

QUALITATIVE (CATEGORICAL) DATA

- **Qualitative data** describes **qualities or characteristics**.
  It is **non-numerical**.

•Gender (Male/Female)
•Blood Group (A, B, AB, O)
•Phone Brand (Samsung, Apple)
•City Name
•Course Name

- You cannot perform calculations like addition or average on this data.

TYPES OF QUALITATIVE DATA

a) Nominal Data

Definition:

- Data with **no natural order**

Examples:

- Gender
- Religion
- Eye color
- Vehicle type

In nominal data No ranking possible.

b) Ordinal Data

Definition:

•Data with a **specific order or ranking**

Examples:

•Feedback: Poor, Average, Good, Excellent

•Education level: School, UG, PG, PhD

•Ranking in class

In ordinal data Order matters

QUANTITATIVE (NUMERICAL) DATA

- **Quantitative data** represents **numbers and quantities**.
  It can be measured and calculated.
- Age
- Height
- Weight
- Marks
- Salary
- Distance
- You can perform calculation over this data

TYPES OF QUANTITATIVE DATA

- **a) Discrete Data**
- **This data include Countable values**
- Usually whole numbers
- **Examples:**
- Number of students in class
- Number of cars in parking
- Number of apps on phone
- You **cannot have fractions**
- (e.g., 2.5 students ❌)

b) Continuous Data

This type of data Can take **any value within a range and** Includes fractions

Examples:

•Height (165.5 cm)

•Weight (60.75 kg)

•Temperature

•Time

Fractions are possible.

CLASS ACTIVITY

Age of students → Quantitative (Continuous)
Height of a person → Quantitative (Continuous)
Weight of a patient → Quantitative (Continuous)
Time taken to complete an exam → Quantitative (Continuous)
Distance from home to college → Quantitative (Continuous)
Favorite subject of students → Qualitative (Nominal)=no fixed change acc. to student
Blood group of patients → Qualitative (Nominal)
Phone brand used by customers → Qualitative (Nominal)
City of residence → Qualitative (Nominal)
Vehicle type (car, bike, bus) → Qualitative (Nominal)

- **Class rank of students** → Qualitative (Ordinal)
- **Customer feedback (Poor, Average, Good, Excellent)** → Qualitative (Ordinal)
- **Education level (School, UG, PG, PhD)** → Qualitative (Ordinal)
- **Performance rating of employees** → Qualitative (Ordinal)
- **Grades in examination (A, B, C, D)** → Qualitative (Ordinal)
- **Number of siblings** → Quantitative (Discrete)
- **Number of students in a classroom** → Quantitative (Discrete)
- **Number of books issued from library** → Quantitative (Discrete)
- **Number of vehicles in parking area** → Quantitative (Discrete)
- **Number of mobile apps installed** → Quantitative (Discrete)

WHY SHOULD WE STUDY DATA VISUALIZATION?

- **Think about this:**
- Why does **Instagram show reels instead of numbers?**
- Why does **Google Maps use colored routes?**
- Why does **COVID data appear as dashboards, not tables?**
- Because **humans understand visuals faster than numbers**
- **Key idea:**
- Data Visualization helps us **see patterns, trends, and insights quickly**.

DATA VISUALIZATION

What Makes Visualization Different from Data?

- **Data Visualization** is the graphical representation of data using charts, graphs, and visuals to understand information easily and quickly.
- The human brain processes images **60,000 times faster** than text.

Data	Visualization
Raw numbers	Meaningful picture
Needs calculation	Needs observation
Time-consuming	Instant understanding

REAL WORLD COMPANIES

- **Real companies using this daily:**
- Google → user behavior
- Netflix → recommendation system
- Amazon → buying patterns
- Government → COVID dashboards

5. MINI CLASSROOM ACTIVITY (5–10 MINUTES)

Activity:

- Ask students:
- How many hours do you spend on social media daily?
- Collect answers and:
- Draw a quick bar chart on the board
- Ask:
  - Who spends the most time?
  - What is the class average?
- Students become part of the data.

DATA VISUALIZATION

- Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps etc.
- Data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

COMMON TYPES OF DATA VISUALIZATION

- There are various types of visualizations where each has a unique purpose in data representation. Here are the most common types:
- **Charts and Graphs:** A chart is a visual representation of data used to summarize and compare information while a graph shows the relationship between variables, usually plotted on X and Y axes. Examples: Bar Charts, Pie Charts, Scatter Plots, Histograms etc.
- **Maps**: They are used to display geographical data which provides spatial context to trends and patterns. Examples: Geographic Maps, Heat Maps.
- **Dashboards:** They combine multiple visualizations into a single interface which provides real-time insights and interactive features for users to explore data.

IMPORTANCE OF DATA VISUALIZATION

- Data visualization is essential for understanding and communicating information effectively. Here are some key reasons why it's important:
- **Simplifies Complex Data:** It turns large and complicated data into visual formats like charts and graphs, making the information easier to understand.
- **Reveals Patterns and Trends**: It helps identify trends, relationships and patterns that are not easily seen in raw data or tables.
- **Saves Time:** Visuals allow quicker interpretation of data, helping users spot key information at a glance instead of manually scanning through numbers.
- **Improves Communication:** It makes it easier to explain data insights to others, especially those who may not be familiar with the technical details.
- **Tells a Clear Story**: Data visuals guide the audience through the information step-by-step, making it easier to reach conclusions and make informed decisions.

REAL-WORLD USE CASES FOR DATA VISUALIZATION

Data visualization is used across various industries to improve decision-making and drive results. Here are a few examples:

- **Business Analytics:** Used to monitor company performance, track KPIs and make data-driven decisions by visualizing trends, sales and customer metrics.
- **Healthcare:** Helps in analyzing patient records, tracking disease outbreaks and managing hospital operations through easy-to-read charts and dashboards.
- **Sports:** Used to visualize player statistics, team performance and match outcomes, helping coaches and analysts improve strategies and training plans.
- **Retail and E-commerce:** Enables tracking of sales, customer preferences and inventory levels, helping businesses adjust stock and marketing efforts effectively.
- ??

CHALLENGES IN DATA VISUALIZATION

- **Data Quality**: Accuracy of visualizations depends on the quality of the data. If the data is inaccurate or incomplete, the insights from the visualization will be misleading.
- **Over-Simplification**: Simplifying data too much can lead to important details being lost like using a pie chart that oversimplifies complex relationships between categories.
- **Choosing the Right Visualization**: Using the wrong type of visualization can distort the message. For example, a pie chart might not work well with many categories which leads to confusion.
- **Overload of Information**: Too much information in a visualization can overwhelm viewers. It's important to focus on key data points and avoid clutter.

BEST PRACTICES FOR EFFECTIVE DATA VISUALIZATION

To ensure our visualizations are impactful and easy to understand we follow these best practices:

- **Audience-Centric Design**: Tailor visualizations to our audience's knowledge. A technical audience may need detailed graphs while a general audience benefits from simpler charts.
- **Design Clarity and Consistency**: Choose the right chart for our data and keep the design clean with consistent colors, fonts and labels and also avoid clutter to ensure clarity.
- **Provide Context**: Always provide context by including labels, titles and data source acknowledgments. This helps viewers to understand the significance of the data and builds trust in the results.
- **Interactive and Accessible Design**: Make visualizations interactive with features like tooltips and filters and ensure accessibility for all users regardless of device or visual needs.

TYPES OF CHARTS

- A chart or graph is a **visual representation of data** that helps us:
  - Understand large datasets quickly
  - Identify patterns and relationships
  - Compare values easily
  - Support data-driven decisions
- The Charts convert **raw numbers into meaningful visuals.**

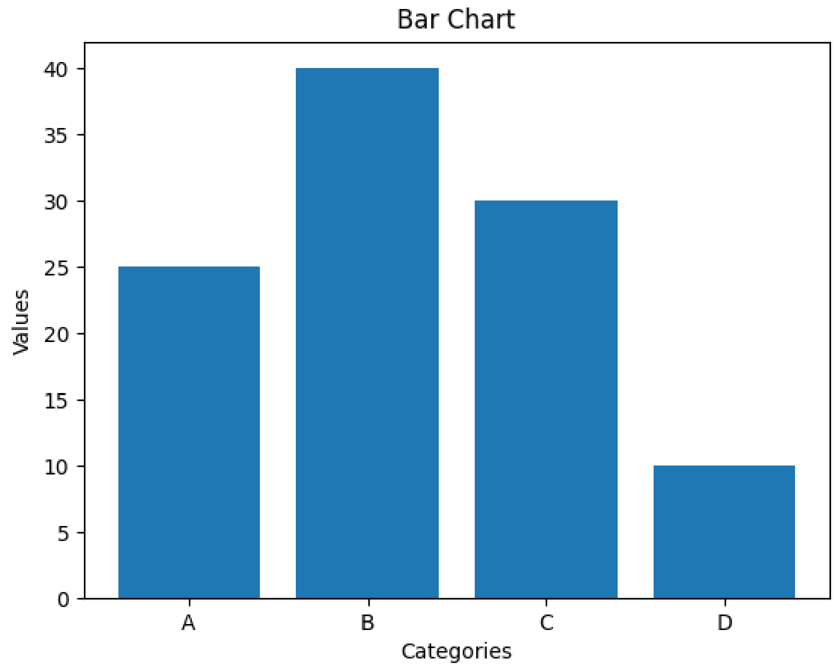
COMMON TYPES OF CHARTS

- Bar Chart
- Line Chart
- Pie Chart
- Histogram
- Scatter Plot
- Area Chart

- Each chart serves a **specific purpose** based on the type of data.

1. BAR CHART

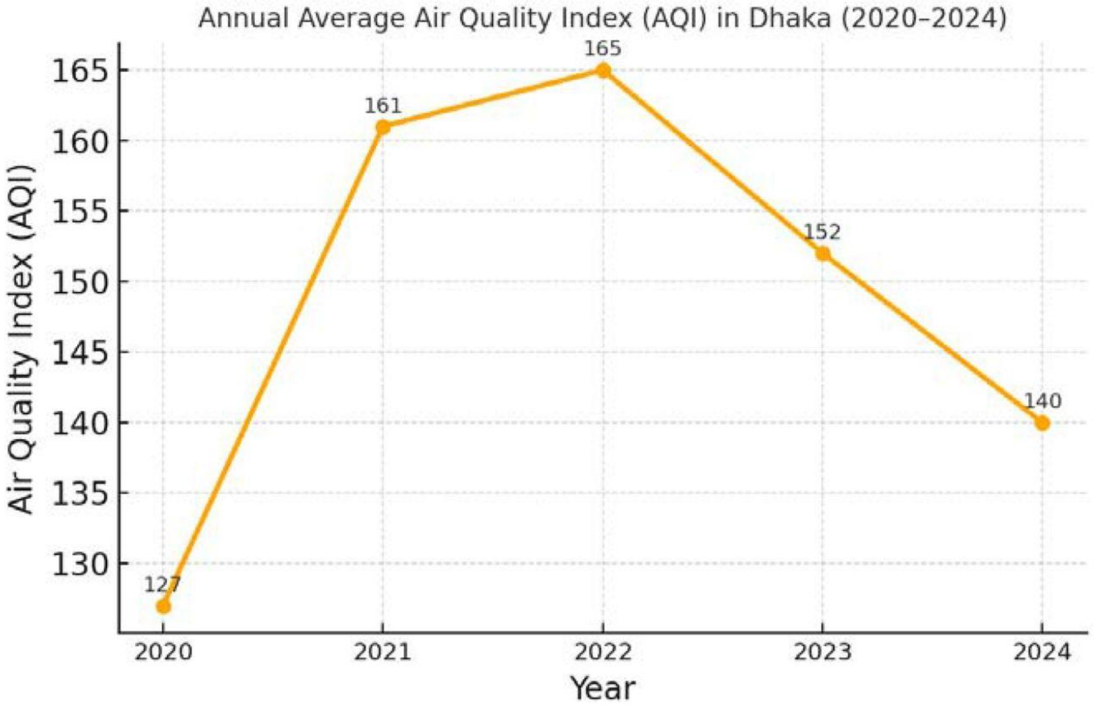
- A bar chart uses **rectangular bars** to represent values for different categories.
- **It is Best Used For**
  - Comparing **categorical data**
  - Showing differences between groups
- **Real-Life Examples**
  - Student marks in different subjects
  - City-wise AQI comparison
  - Product-wise sales in a store
- **Key Advantage**
- Easy comparison between categories.



2. LINE CHART

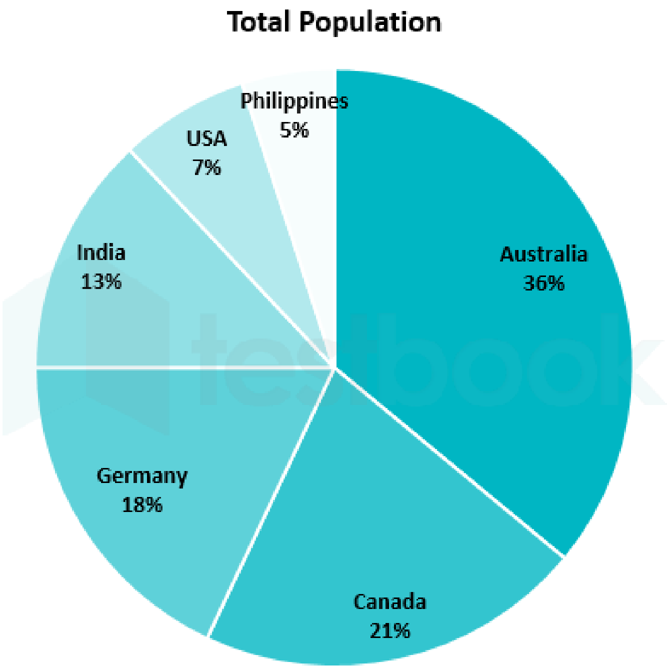
- A line chart displays data points connected by lines to show **changes over time**.
- **Best Used For**
  - Time-series data
  - Trend analysis
- **Real-Life Examples**
  - Monthly sales growth
  - AQI variation over days
  - Temperature changes during a day
- **Key Advantage**

Clearly shows trends and patterns.



3. PIE CHART

- A pie chart is a circular chart divided into slices representing **parts of a whole (100%)**.
- **Best Used For**
  - Showing percentages
  - Part-to-whole relationship
- **Real-Life Examples**
  - Grade distribution of students
  - Monthly household expenses
  - Market share of companies
- **Key Advantage**
- Simple visual understanding of proportions.



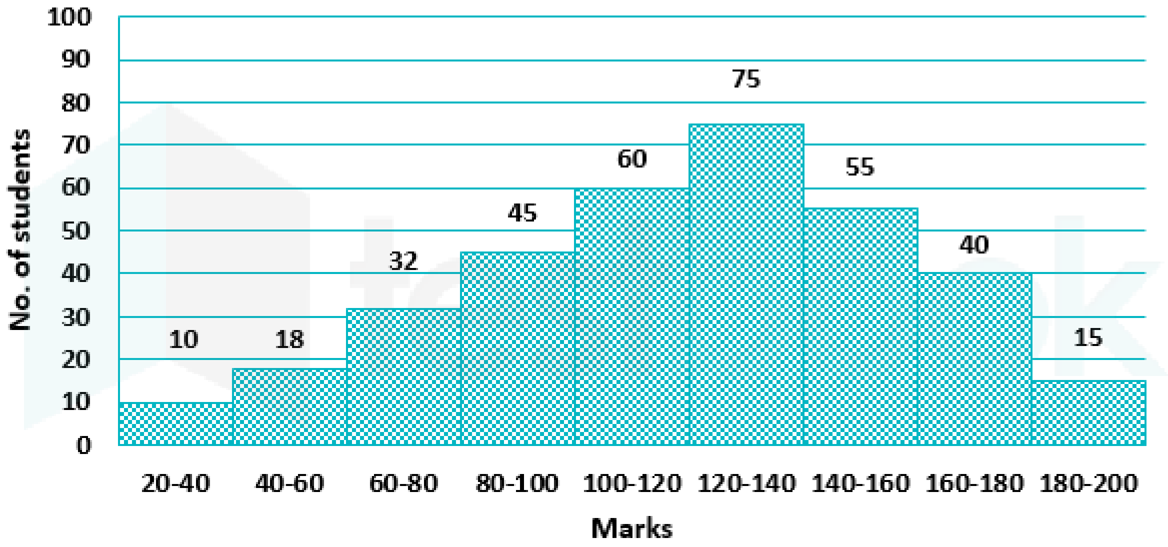
4. HISTOGRAM

- A histogram looks similar to a bar chart but represents **frequency distribution of numerical data**.
- **Important Difference:** Bar chart → categorical data while
- Histogram → continuous numerical data
- **Best Used For**

Understanding data distribution

Identifying patterns such as skewness or normal distribution

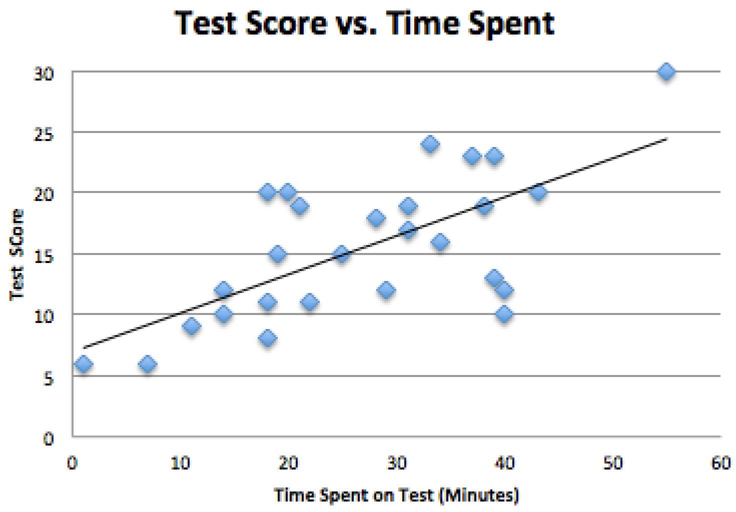
- **Real-Life Examples**
  - Distribution of student marks
  - Age distribution of a population
  - Distribution of daily AQI values



- **Key Advantage:** Shows how data values are spread across ranges.
- **Limitation:** Not suitable for categorical data comparison.

5. SCATTER PLOT

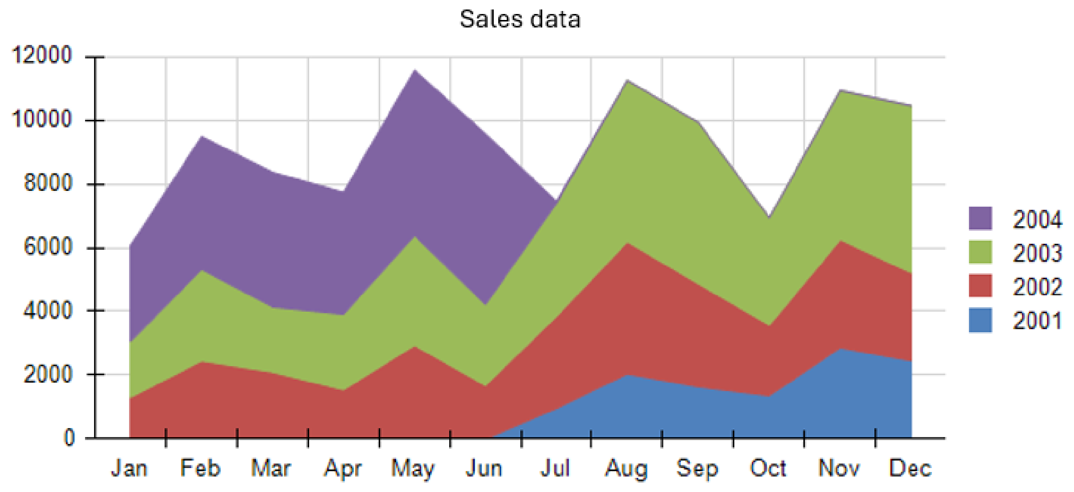
- A scatter plot displays individual data points using dots to show the **relationship between two numerical variables**.
- **Best Used For**
  - Identifying relationships or correlations
  - Observing patterns and outliers
- **Real-Life Examples**
  - Study hours vs exam marks
  - Advertising cost vs sales revenue
  - Temperature vs electricity consumption



- **Key Advantage:** Helps identify correlation (positive, negative, or none).
- **Limitation:** Not useful for showing totals or trends over time.

6. AREA CHART

- An area chart is similar to a line chart but the area below the line is **filled with color**.
- **Best Used For**
  - Showing cumulative values over time
  - Comparing multiple quantities
- **Real-Life Examples**
  - Website traffic growth over months
  - Electricity consumption over time
  - Sales contribution of multiple products
- **Key Advantage:** Shows both **trend and volume** clearly.
- **Limitation:** Too many overlapping areas can cause confusion.



CHOOSING THE RIGHT VISUALIZATION FOR DATA

- Selecting right chart is key to effective data visualization.
- Different chart types highlight different patterns, trends or comparisons. Before choosing the right chart it's essential to understand the nature of your data and the message you want to convey.
- To understand consider the following questions:
  - What is the primary purpose of the visualization?
  - What type of data are you working with?
  - How many variables are you visualizing?
  - What is the volume of data points?

KEY POINTS FOR CHOOSING THE RIGHT VISUALIZATION

Choosing the correct visualization is important because the **right chart makes data easy to understand**, while the **wrong chart can confuse or mislead**.

1. Understand the Type of Data

- The first step is to identify what kind of data you have.
- **Categorical data** →
  Example: Subjects, cities, products
- **Numerical data** →
  Example: Marks, age, temperature
- **Time-series data** →
  Example: Monthly sales, AQI over time

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- **2. Identify the Purpose of Visualization**
- Decide what you want to show.
- **Comparison** → Bar chart
  Example:
- Marks of students in different subjects
- **Trend over time** → Line chart
  Example: Sales growth per month
- **Proportion or percentage** → Pie chart
  Example: Grade distribution

3. Number of Data Categories
The number of values affects chart selection.
Few categories → Pie chart
Many categories → Bar chart or Histogram
Example:
5 products → Pie chart
20 products → Bar chart

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- **4. Relationship Between Variables**
- If you want to show how two variables are related:
- **Use Scatter Plot**
  Example: Study hours vs exam marks

- **5. Distribution of Data**
- To understand how data values are spread:
- **Use Histogram**
  Example: Distribution of student marks

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- **6. Avoid Misleading Visuals**
- Wrong charts can give false interpretation.
  - ➢ Avoid pie charts with too many slices
  - ➢ Avoid 3D charts that distort values
- *Example:* A 3D pie chart may exaggerate differences.
- **7. Keep Visualization Simple**
  - Simple visuals are more effective.
  - Use limited colors
  - Clear labels and titles
  - Avoid unnecessary decoration